

A Note on Prime Avoidance in Reflected Sieve Sets

Abstract

We study reflected prime structures through the mirror map $x \mapsto N - x$, with particular attention to primes in short intervals near the center. We show several structures in the local distributions of primes and establish that either reflected prime pairs exist for every even integer, or a rigid arithmetic structure imposed by mirror obstruction classes.

1 Introduction

Let n be a large integer and consider representations of $2n$ as a sum of two primes. Writing such a representation as

$$2n = p + q, \quad p \leq q,$$

this is equivalent to the existence of a prime $p < n$ such that $2n - p$ is also prime. We refer to $(p, 2n - p)$ as an *equidistant prime pair* centered at n .

The goal of this note is not to prove Goldbach's conjecture, but to give structural evidence for its truth by showing that any counterexample would force a strong and unexpected arithmetic rigidity.

“tex

Lemma 1 (Fixed-level survivor criterion). *Let $I(n) = (n - L, n]$, and for $\alpha > 0$ define*

$$S_\alpha(n) := \{x \in I(n) : x \not\equiv 2n \pmod{q} \text{ for every prime } q \leq n^\alpha\}.$$

If $S_\alpha(n) \cap \mathbb{P} \neq \emptyset$ for some $\alpha > 1/2$, then $2n$ admits as the sum of two primes.

Remark 1. Thus the fixed-level sieve gives a sufficient local criterion: a single prime survivor past the square-root scale yields a reflected prime pair.

Theorem 1 (Mirror-cover equivalence). *Let $N \geq 4$, and let $X \subseteq \{x \in \mathbb{Z} : 2 \leq x < N - 1\}$. For each $x \in X$, define its reflection by $\rho_N(x) := N - x$. Define the mirror-captured subset of X by*

$$\mathcal{C}_N(X) := \left\{ x \in X : \exists \text{ prime } q \leq \sqrt{N - x} \text{ such that } x \equiv N \pmod{q} \right\}.$$

Then $X \setminus \mathcal{C}_N(X) = \{x \in X : N - x \in \mathbb{P}\}$. Equivalently, $X \not\subseteq \mathcal{C}_N(X)$ if and only if there exists $x \in X$ whose reflection $N - x$ is prime.

Lemma 2 (Pointwise mirror obstruction). *Let $2n$ be an even integer, and let $I(n) = (n - L, n]$. For each prime q , define the mirror obstruction class*

$$K_q(n) := \{p \in I(n) \cap \mathbb{P} : p \equiv 2n \pmod{q}\}.$$

Define the pointwise captured subset of $I(n) \cap \mathbb{P}$ by

$$\mathcal{C}_I(n) := \left\{ p \in I(n) \cap \mathbb{P} : \exists \text{ prime } q \leq \sqrt{2n - p} \text{ such that } p \equiv 2n \pmod{q} \right\}.$$

Suppose that no prime $p \in I(n) \cap \mathbb{P}$ has prime reflection $2n - p$. Then $I(n) \cap \mathbb{P} \subseteq \mathcal{C}_I(n)$. Equivalently, for every $p \in I(n) \cap \mathbb{P}$, there exists a prime $q \leq \sqrt{2n - p}$ such that $q \mid (2n - p)$.

Lemma 2 identifies the rigid arithmetic alternative to a reflected prime pair: under local failure, the entire prime set $I(n) \cap \mathbb{P}$ must be captured by mirror obstruction classes. This is a total-capture condition on a local prime set of natural size

$$\#(I(n) \cap \mathbb{P}) \approx \frac{L}{\log n}.$$

Thus a local Goldbach failure would not merely remove a few candidates; it would require every prime $p \in I(n) \cap \mathbb{P}$ to have composite reflection $2n - p$, and hence to lie in some center-dependent class

$$K_q(n) = \{p \in I(n) \cap \mathbb{P} : p \equiv 2n \pmod{q}\}, \quad q \leq \sqrt{2n}.$$

The shortness of $I(n)$ also restricts how such classes can be reused: if $p_1, p_2 \in K_q(n)$ with $p_1 \neq p_2$, then $q \mid (p_1 - p_2)$, while $|p_1 - p_2| < L$, so $q \leq L$. Hence every modulus $q > L$ can capture at most one local prime. Consequently, any total covering of $I(n) \cap \mathbb{P}$ would require concentrated reuse among the small moduli $q \leq L$.

Remark 2 (Goldbach equivalence). Theorem 1 gives the obstruction form of Goldbach in the interval $I(n)$. The existence of a Goldbach representation with first prime in $I(n)$ is equivalent to the failure of total capture by the mirror obstruction classes:

$$\exists p \in I(n) \cap \mathbb{P} \text{ such that } 2n - p \in \mathbb{P} \iff I(n) \cap \mathbb{P} \not\subseteq \bigcup_{\substack{q \leq \sqrt{2n-p} \\ q \text{ prime}}} K_q(n).$$

Thus, for the purposes of proving Goldbach for $2n$, it is enough to show that at least one prime in the local interval survives all center-dependent mirror obstructions.

This is a different formulation from the standard global additive viewpoint. The Hardy–Littlewood circle method frames Goldbach through the representation count $R(2n)$ [4], while sieve-theoretic approaches, such as Chen’s theorem [3], replace the exact two-prime condition by the relaxed target $2n = p + P_2$, where p is prime and P_2 has at most two prime factors. Exceptional-set results, beginning with Montgomery and Vaughan [1], instead measure the size of the set of even integers for which the available methods do not produce a Goldbach representation.

By contrast, the mirror formulation keeps the exact two-prime condition and changes the object of study: it turns Goldbach in $I(n)$ into the local intersection problem

$$(I(n) \cap \mathbb{P}) \cap \{x \in I(n) : 2n - x \in \mathbb{P}\} \neq \emptyset.$$

Here, failure is no longer an undifferentiated absence of representations; it becomes the concrete assertion that the local prime set $I(n) \cap \mathbb{P}$ is totally captured by center-dependent congruence obstructions.

Remark 3 (Mirror intersection number). Define $P_I(n) := I(n) \cap \mathbb{P}$, $R_I(n) := \{x \in I(n) : 2n - x \in \mathbb{P}\}$, and the mirror intersection number $M_I(n) := \#(P_I(n) \cap R_I(n))$. Thus the local problem is the positivity question $M_I(n) > 0$. Consequently, any resolution of Goldbach also determines a structural feature of the reflected-prime structure encoded by these mirror intersection numbers

We now introduce the minimal local notation needed for the argument. For each center n , the mirror obstruction classes $K_q(n)$ record the congruence obstructions to the primality of a reflected value $2n - p$. The only cutoff used is the square-root cutoff for the reflected value: at this cutoff, escaping all mirror obstructions is equivalent to having prime reflection. Hence

either some local prime survives, giving a reflected prime pair, or every local prime is captured, so that each reflection $2n - p$ is forced composite by a center-dependent congruence obstruction. This is the rigid total-capture alternative.

2 Mirror Obstruction Classes and a Local Reformulation

Fix n large, and define the short interval

$$I(n) := (n - L, n], \quad L := n^{0.6}.$$

For each prime $p \in I(n) \cap \mathbb{P}$, its reflection across the center n is $2n - p$.

Thus, $2n$ admits as the sum of two primes if a prime $p \in I(n) \cap \mathbb{P}$ whose reflection $2n - p$ is also prime.

For each prime q , define the *mirror obstruction class*

$$K_q(n) := \{p \in I(n) \cap \mathbb{P} : p \equiv 2n \pmod{q}\}.$$

Equivalently, $p \in K_q(n)$ if and only if

$$q \mid (2n - p).$$

Thus $K_q(n)$ consists of those primes in $I(n)$ whose reflections are eliminated by divisibility by q .

3 Pointwise Mirror Survivors

The cutoff in the mirror problem is not a fixed sieve level B , but the square-root cutoff attached to each reflected value. For $x \in I(n)$, define

$$B_x(n) := \sqrt{2n - x}.$$

We say that x survives the pointwise mirror sieve if $q \nmid (2n - x)$ for every prime $q \leq B_x(n)$. By the elementary divisor criterion,

$$x \text{ survives the pointwise mirror sieve} \iff 2n - x \in \mathbb{P}.$$

Thus the pointwise survivor set is exactly the reflected-prime set

$$R_I(n) := \{x \in I(n) : 2n - x \in \mathbb{P}\}.$$

4 Proof of Theroem 1 and Lemma 1

Lemma 1 proof:

Proof. Let $p \in S_\alpha(n) \cap \mathbb{P}$. Then no prime $q \leq n^\alpha$ divides $2n - p$. Since $p \in I(n) = (n - L, n]$, we have $n \leq 2n - p < n + L$. For $\alpha > 1/2$ and $L = n^{0.6}$, we have, for n sufficiently large, $\sqrt{2n - p} \leq \sqrt{n + L} < n^\alpha$. Thus $2n - p$ has no prime divisor at most $\sqrt{2n - p}$. Hence $2n - p$ is prime. Since p is prime, $2n = p + (2n - p)$ admits as the sum of two primes. $\square$

proof of theroem 1:

Proof. For $x \in X$, the congruence $x \equiv N \pmod{q}$ is equivalent to $q \mid (N - x)$. Thus $x \in C_N(X)$ exactly when the reflection $N - x$ has a prime divisor $q \leq \sqrt{N - x}$.

If $N - x$ is composite, then it has a prime divisor $q \leq \sqrt{N - x}$, so $x \in C_N(X)$. Conversely, if $x \in C_N(X)$, then $N - x$ has a prime divisor $q \leq \sqrt{N - x}$. Since $N - x > 1$, this implies $N - x$ is composite. Therefore $x \notin C_N(X)$ if and only if $N - x \in \mathbb{P}$. Hence

$$X \setminus C_N(X) = \{x \in X : N - x \in \mathbb{P}\}.$$

The final equivalence follows. $\square$

Proof. Let $p \in I(n) \cap \mathbb{P}$. By hypothesis, $2n - p$ is composite, so some prime $q \leq \sqrt{2n - p}$ divides $2n - p$. Equivalently, $p \equiv 2n \pmod{q}$, and hence $p \in C_I(n)$. Since $p \in I(n) \cap \mathbb{P}$ was arbitrary, we have $I(n) \cap \mathbb{P} \subseteq C_I(n)$. $\square$

References

- [1] H. L. Montgomery and R. C. Vaughan, *The exceptional set in Goldbach's problem*, Acta Arithmetica **27** (1975), 353–370.
- [2] T. Oliveira e Silva, S. Herzog, and S. Pardi, *Empirical verification of the even Goldbach conjecture and computation of prime gaps up to $4 \cdot 10^{18}$* , Mathematics of Computation **83** (2014), no. 288, 2033–2060.
- [3] J. Chen, *On the representation of a large even integer as the sum of a prime and the product of at most two primes*, Scientia Sinica **16** (1973).
- [4] H. A. Helfgott, *The Ternary Goldbach Problem*, Annals of Mathematics Studies, 2015.